Fermat's Little Theorem

An End-to-End Meta-Math Walkthrough

Dibakar Sigdel[1]

[1] *Mindverse Computing LLC, Lynnwood, WA 98087, USA*

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Abstract

Multiplication by a nonzero residue permutes a finite set. Comparing its products gives the desired power, provided the factorial is proved invertible before cancellation. This report presents a complete proof relative to explicit imports, a dependency graph, versioned node contracts, five-component activities, exact examples, and an illustrative repair. The interactions are authored explanations, not a transcript of deployed agents or a proof-assistant certificate.

Project NT-04 Report version 1 Number Theory

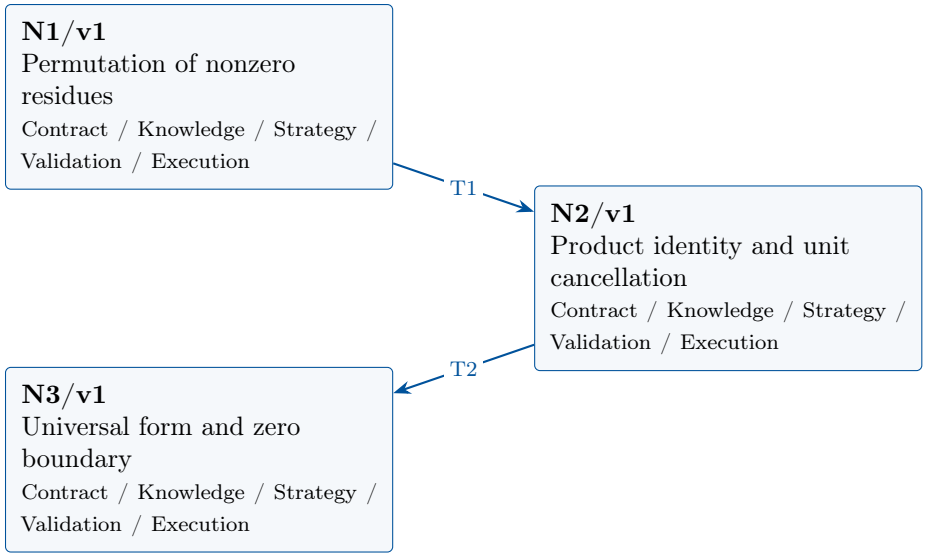
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For every prime p and every integer a, $a^p \equiv a \pmod{p}$. Equivalently, for $p \nmid a$, one has $a^{p-1} \equiv 1 \pmod{p}$.*

A. Foundations and permitted imports

1. NT-01/N5, v1: a residue modulo $m \geq 2$ is invertible iff it is coprime to m. In particular every nonzero residue modulo a prime p is a unit.
2. NT-02/N2, v1: a prime dividing a product divides a factor. An injective map from a finite set to itself is bijective; finite products in the commutative residue ring are unchanged by permutation. No group-order theorem is imported.

II. NODE INTERACTION WORKFLOW



Blue arrows are accepted mathematical dependencies, with exact transfers T1–T2 below. Repair messages are described separately; they are not premises of the proof.

A. Versioned evidence-transfer ledger

T1: N1/E1 v1 to N2.1
Claim. The map $[j] \mapsto [aj]$ permutes the nonzero classes $1, \ldots, p-1$ modulo p.
Compatibility. Products run through the same finite set of nonzero classes.
Evidence. Node N1 activities below and the continuous proof in Section IV.

T2: N2/E2 v1 to N3.1
Claim. If $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.
Compatibility. Apply the power congruence only when *pmida*.
Evidence. Node N2 activities below and the continuous proof in Section IV.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Permutation of nonzero residues

Object and role. Transport a modular multiplication problem into a finite permutation.

Inputs. p prime, $p \nmid a$; NT-01 inverse criterion and NT-02 Euclid lemma.

C — Contract

Publish a bijection only on the nonzero residue classes.

K — Knowledge

Retain a certified inverse of a modulo p and the finite set of $p-1$ classes.

S — Strategy

Prove the map lands in the set and is injective; finiteness supplies surjectivity.

V — Validation

A nonzero representative modulo a composite need not be a unit.

E — Execution

If $p \mid aj$ with $1 \le j < p$, Euclid's lemma would give $p \mid j$, a contradiction. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N1.1 If $p \mid aj$ with $1 \le j < p$, Euclid's lemma would give $p \mid j$, a contradiction.

N1.2 From $aj \equiv ak \pmod{p}$, multiply by a^{-1} to get $j \equiv k \pmod{p}$.

N1.3 The finite self-map is injective and hence a permutation.

Export E1/v1. The map $[j] \mapsto [aj]$ permutes the nonzero classes $1, \dots, p-1$ modulo p.

Acceptance gate. The range contains only nonzero classes and every input has a distinct image.

Targeted repair. If $p \mid a$, route to the zero case in N3; do not pass a permutation export to N2.

B. N2/v1: Product identity and unit cancellation

Object and role. Extract the exponent from equal products and justify the cancellation factor.

Inputs. N1/E1; primality of p; NT-01 and NT-02.

C — Contract

Publish the power congruence only after certifying the factorial as a unit.

K — Knowledge

Retain the permutation and the exact product $P = (p-1)!$.

S — Strategy

Compare products before and after multiplication, then multiply by a certified inverse of P.

V — Validation

Do not cancel a merely nonzero integer expression in an arbitrary residue ring.

E — Execution

The permutation yields $a^{p-1}P \equiv P \pmod{p}$. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N2.1 The permutation yields $a^{p-1}P \equiv P \pmod{p}$.

N2.2 No factor $1 \le j < p$ is divisible by p, so Euclid's lemma gives $p \nmid P$.

N2.3 Use the inverse of P to conclude $a^{p-1} \equiv 1 \pmod{p}$.

Export E2/v1. If $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

Acceptance gate. The equal products and the unit certificate for P are both present; $p = 2$ uses $P = 1$.

Targeted repair. If the factorial unit claim fails, revisit the prime modulus and invalidate the cancelled conclusion.

C. N3/v1: Universal form and zero boundary

Object and role. Join zero and nonzero cases and prove the stated equivalence.

Inputs. N2/E2 for $p \nmid a$; arbitrary integer a.

C — Contract

Publish the universal claim only after both residue cases are covered.

K — Knowledge

Retain the nonzero-case scope and the inverse of a only where it exists.

S — Strategy

Multiply by a in the nonzero case and prove the zero case directly.

V — Validation

Do not divide the universal congruence by a when $p \mid a$.

E — Execution

If $p \nmid a$, multiply N2's congruence by a. Continue with the numbered deductions below, retaining the exact export.

Numbered mathematical activities

N3.1 If $p \nmid a$, multiply N2's congruence by a.

N3.2 If $p \mid a$, both a^p and a are zero modulo p.

N3.3 Conversely for $p \nmid a$, multiply $a^p \equiv a$ by a^{-1}.

Export E3/v1. $a^p \equiv a \pmod{p}$ for every integer a, with the equivalent nonzero formulation.

Acceptance gate. Every integer lies in exactly one case and the reverse implication is restricted to units.

Targeted repair. A proposed nonzero formulation for multiples of p is rejected; preserve the valid universal form.

IV. CONTINUOUS MATHEMATICAL PROOF

Proof. Assume first $p \nmid a$. The map $[j] \mapsto [aj]$ on the nonzero residue classes modulo p is well-defined and has nonzero values: if $p \mid aj$, Euclid's lemma and $p \nmid a$ would force $p \mid j$. Moreover, a has an inverse by NT-01, so $aj \equiv ak$ implies $j \equiv k$. This is an injective self-map of a finite set, hence a permutation (N1).

Multiplying all its values and comparing with the original product gives (N2)

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Each factor $1, \ldots, p-1$ is not divisible by p. Repeated Euclid's lemma implies $p \nmid (p-1)!$, so its gcd with p is 1. The inverse criterion therefore justifies multiplying by the inverse of the factorial. This yields $a^{p-1} \equiv 1 \pmod{p}$. For $p = 2$, the product is simply 1, so the same argument applies.

Multiplying the nonzero-case congruence by a yields $a^p \equiv a$. If $p \mid a$, both sides are zero modulo p, proving the universal formulation (N3). Conversely, for $p \nmid a$, multiply the universal formulation by the inverse of a to recover $a^{p-1} \equiv 1$. Residue classes cover negative as well as nonnegative integers.

□

V. WORKED ORCHESTRATION

For $a = 3, p = 7$, N1 constructs the permutation

$$1, 2, 3, 4, 5, 6 \longmapsto 3, 6, 2, 5, 1, 4 \pmod{7}.$$

N2 uses $6! = 720 \equiv 6 \pmod{7}$, whose inverse is 6, to cancel the product. Thus $3^6 = 729 = 104 \cdot 7 + 1$. Multiplying by 3 gives $3^7 \equiv 3 \pmod{7}$. For $a = 14$, N3 instead uses the zero case $14^7 \equiv 14 \equiv 0 \pmod{7}$; it does not assert $14^6 \equiv 1$.

A. Boundary and hypothesis audit

- At $p = 2$, odd a satisfies $a \equiv 1 \pmod{2}$, while even a uses the zero case.
- Primality cannot simply be dropped: $2^4 \equiv 0 \not\equiv 2 \pmod{4}$. Even coprimality alone does not preserve the exponent $m - 1$: $3^3 \equiv 3 \not\equiv 1 \pmod{4}$.
- Modular cancellation can fail: $2 \cdot 1 \equiv 2 \cdot 4 \pmod{6}$, but $1 \not\equiv 4 \pmod{6}$.

VI. ILLUSTRATIVE FAILURE AND REPAIR

This is a deliberately authored teaching trace. A diagnostic asks for correction; it cannot prove a predecessor by importing its descendant's conclusion. A revised export must be checked before its consumers are reused.

Candidate. N2 cancels a factorial modulo an arbitrary positive modulus merely because the factorial is a nonzero integer.

Detection. Validation supplies $2 \cdot 1 \equiv 2 \cdot 4 \pmod{6}$; the common factor 2 is not a unit.

Correction. Knowledge retrieves the inverse criterion. Under the actual prime-modulus contract, Execution proves $p \nmid (p-1)!$ before cancellation.

Recheck. Recompute N2's product deduction and N3's nonzero branch; the zero branch remains a separate direct argument.

Publication. Contract exports Fermat's theorem for prime moduli. A composite-modulus result would require a different exponent and proof.

VII. REUSABLE RESULT AND PROVENANCE

Reusable exports retain each node's version 1 statement, assumptions, and proof. The final export is the theorem in Section 1, proved relative to its declared imports.

Source record. Adapted from the original project Markdown:
`fermat-s-little-theorem.md`

Typography, workflow conventions, and component cards follow the workspace Meta-Math manuscript. This authored exposition does not claim formal certification.